



Welcome 😊!!

Agenda

↳ Variables

↳ Input

variable

↳

int a = 10;

10
a

X Undeclared Variable

X Re declaration

✓ Re used

Vedam
School of Technology



Quiz Time

<https://ahaslides.com/MXJKG>

Join with your Name and registered email

Format

Name <email>

Example

Aman <aman@gmail.com>

Vedam
School of Technology



Quiz 1 :

What will happen?

```
int abc = 10;
```

```
System.out.print(ABC);
```

↑
11102

Quiz 2 :

What will happen?

```
System.out.print(n); → 11102
```

```
int n = 10;
```

Quiz 3:

What will happen?

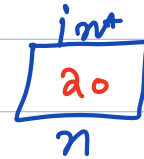
int n = 10; ←

int n = 20; — error

System.out.print(n);



int n = 10;
n = 20;



Quiz 4:

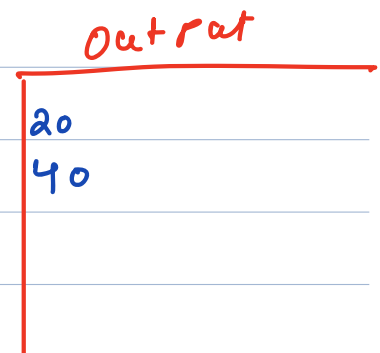
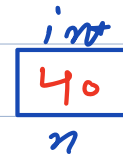
What will be the output?

int n = 20; ←

System.out.print(n); →

n = 40;

System.out.print(n);





Quiz 5:

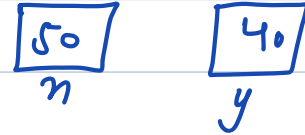
What will be the output?

```
int n = 20;
```

```
int y = 40;
```

```
n = y + 10;
```

```
System.out.print(n);
```



→ 50

Quiz 6:

What will happen?

```
int n = 20, y = 40;
```

```
System.out.print(n + y);
```

error

Input

```
import java.util.Scanner;
```

```
Scanner scn = new Scanner ( System.in );
```

```
int a = scn.nextInt();
```

```
System.out.print (a);
```

20
a



Quiz 7:

Input: 30

```
scanner scn = new scanner(system.in);
```

```
int x = scn.nextInt();
```

↑
error

```
System.out.println(x);
```

Quiz 8:

Input: 30

```
Scanner scn = new Scanner(System.in);
```

```
int x = scn.nextInt();
```

```
System.out.println(x);
```

↑
20



Quiz 9:

Input: 24

```
Scanner scn = new Scanner(System.in);
```

```
int x = scn.nextInt();
```

```
int y = scn.nextInt();
```

```
System.out.println(x);
```

24
x

y

Quiz 10:

Input: 24 32

```
Scanner scn = new Scanner(System.in);
```

```
int x = scn.nextInt();
```

```
int y = scn.nextInt();
```

```
System.out.println(x);
```

24

24
x

32
y



Quiz 11:

What is the correct syntax to declare a variable in Java?

type variableName = value; ✓

variableName type = value; ✗

variableName = value; ✗

None of the above ✗

Quiz 12:

Declare an integer variable with the name y and value 20.

`int y = 20;`



Quiz 13:

What will be the output?

```
int a = 10;
```

10
a

```
int b = 20;
```

20
b

```
System.out.println(a + b);
```

30

Quiz 14:

What will be the output?

```
int a = 20;
```

```
int b = 30;
```

```
System.out.println(a + " " + b);
```

20 30



Quiz 15:

What will happen?

```
int temp = 10;
```

```
System.out.println(Temp);
```

↑
error

Quiz 16:

What will happen?

```
System.out.println(a);
```

```
int a = 10;
```

↑
error



Quiz 17:

What will happen?

```
int a = 20;
```

```
int a = 30; ← error
```

```
System.out.println(a);
```

Quiz 18:

What will be the output?

```
int a = 10;
```

```
int b = 20;
```

```
a = 20b + 30; →
```

```
System.out.println(a);
```

↑
50

50
a

20
b

Variable Naming Rules

1. Name can contain lowercase, uppercase or digits [0-9] or \$ [dollar] or '_' (Underscore)

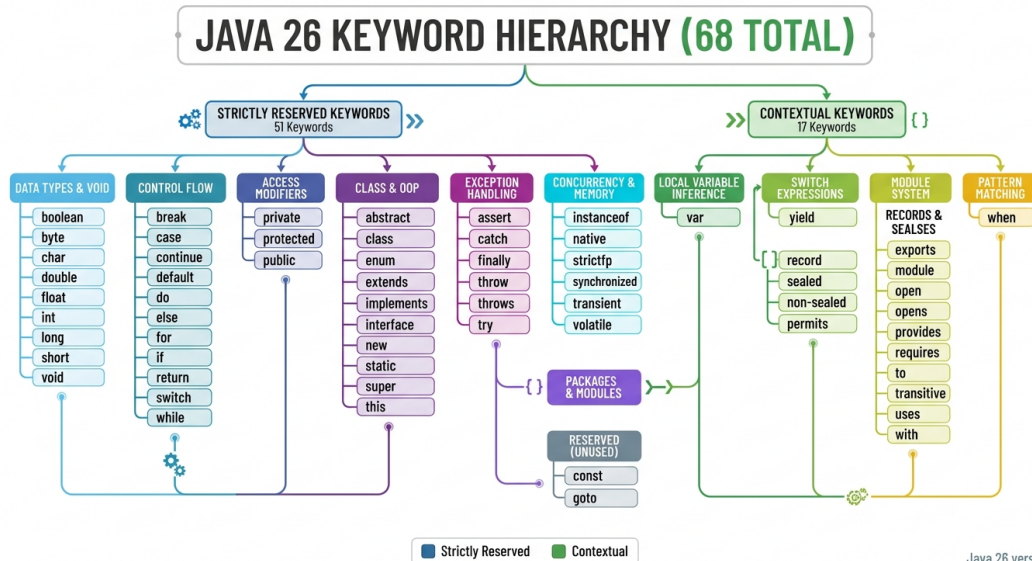
~~X @ / " ' , * , + , ! , ? X~~

2. First letter of the variable name can't be number

e.g. ~~9a~~ a9 ✓ a99, c2134

3. We can't use reserved words as variable name.

Reserved words : words which are predefined in java.



meet back
@ 12:30



Quiz 19:

How many of the following are valid variable names?

int a = 10; ✓

int 1y = 20; ✗

int y@b = 30; ✗

1

Vedam
School of Technology

Quiz 20:

How many of the following are valid variable names?

int _y = 10; ✓

int any = 20; ✓

int n a = 30; ✗

int yf2 = 45; ✓

3



Quiz 21:

What will happen?

```
int static = 60;
```

error

```
System.out.println(static);
```

Different Categories of data

a) Number

Integer decimal

byte $\rightarrow -2^7$ to $2^7-1 \rightarrow 8$

short $\rightarrow -2^{15}$ to $2^{15}-1 \rightarrow 16$

int $\rightarrow -2^{31}$ to $2^{31}-1 \rightarrow 32 \rightarrow 10^9$

long $\rightarrow -2^{63}$ to $2^{63}-1 \rightarrow 64 \rightarrow 10^{18}$

b) Text

letter words
character string

c) boolean

$\rightarrow$ True
 $\rightarrow$ false

$2^{10} \rightarrow 1024 \rightarrow 10^3$



5L



10L



25L



50L

water

4L

✓

✓

✓

✓

15L

X

X

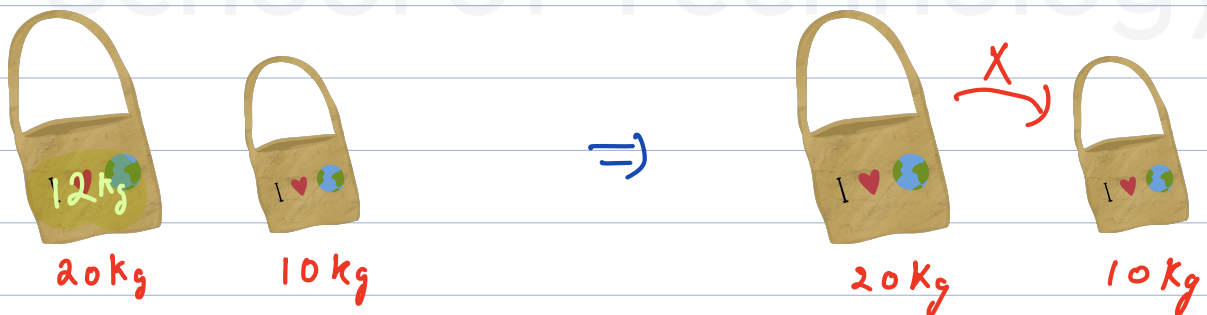
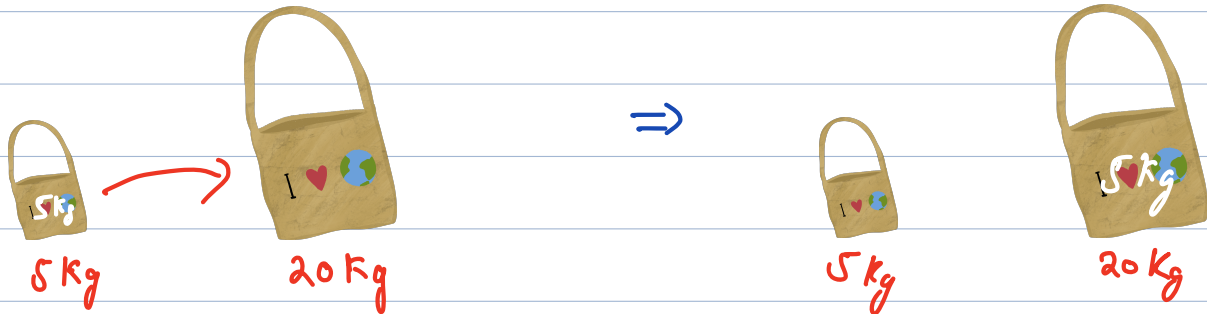
✓

✓

Name	Width	Range
long	64	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
int	32	-2,147,483,648 to 2,147,483,647
short	16	-32,768 to 32,767
byte	8	-128 to 127

Type Casting

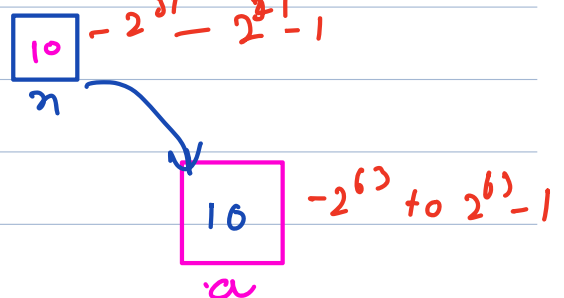
converting one type to other



Small
to
big

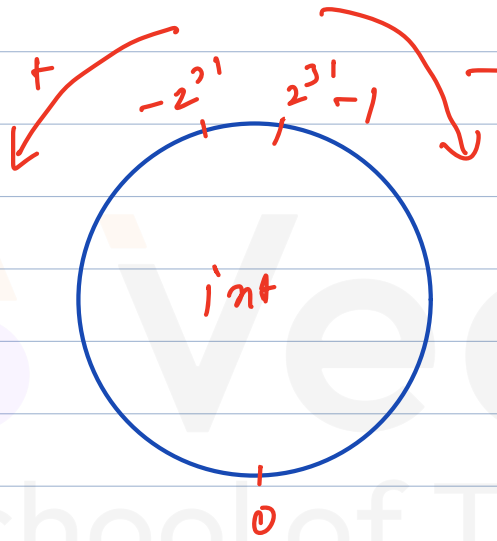
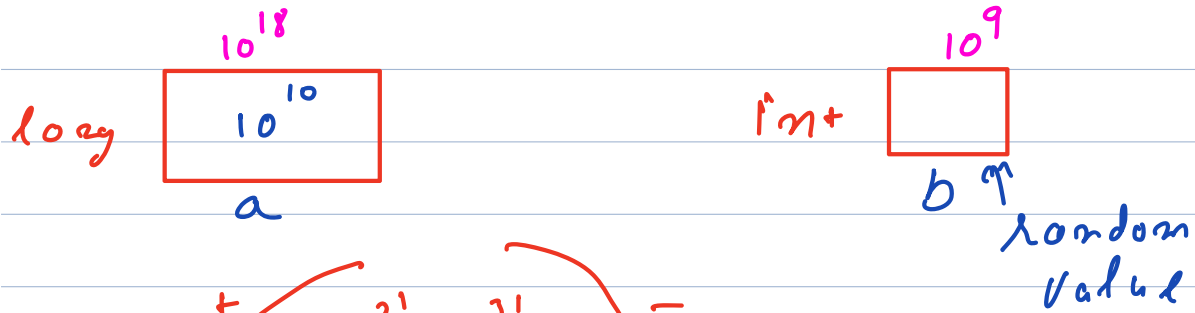
```
int n = 10;  
long a = n;  
↳ implicit  
Type casting
```

loopy conversion





big long a = 100000;
to
small int b = (int)a ; ✓✓ } explicit
Type casting



Vedam
School of Technology

Quiz 22 :

What will be the output?

```
int a = 10000;
```

```
long b = a;
```

```
System.out.println(b);
```

↑
10000

small
+
big

implicit

Quiz 23:

What will happen?

```
long n = 100000;
```

```
int y = n; → error
```

big
+
small

Explicit

```
System.out.println(y);
```

int y = (int) n;